UNIVERSITY OF ENGINEERING & TECHNOLOGY

PESHAWAR

Computer Programming

LABORATORY REPORT

Lab No. 05

Topic:

The while Loop in C++

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Objectives

The objectives of this laboratory session are to learn and apply the following C++ loop constructs:

- • Understand and implement the while loop for repetitive execution.
- • Understand and implement nested while loops.
- • Understand and implement the do...while loop and distinguish it from the while loop.
- • Apply loop constructs to solve practical programming problems.

Theory

The while Loop

A while loop is a control flow statement that allows code to be executed repeatedly based on a given condition. The while loop consists of a block of code and a condition. The condition is evaluated first, and if the condition is true, the code within the block is executed. This process repeats until the condition becomes false.

The general syntax of the while loop is:

```
while (condition)
{
    // statements to execute
}
```

The flow of execution is as follows:

Step 1: The condition is evaluated.

Step 2: If the condition is true, the loop body executes.

Step 3: The condition is evaluated again.

Step 4: Steps 2-3 repeat until the condition becomes false.

Nested while Loop

A nested while loop is a while loop placed inside another while loop. For each iteration of the outer loop, the inner loop completes all of its iterations. This construct is useful for working with multi-dimensional data structures, patterns, and complex iteration logic.

The general syntax of nested while loops is:

```
while (outer_condition)
{
    while (inner_condition)
    {
        // inner loop statements
    }
    // outer loop statements
}
```

The do...while Loop

The do...while loop is a variant of the while loop with one important difference: the body of the do...while loop is executed once before the condition is checked. This guarantees that the loop body executes at least one time, regardless of whether the condition is initially true or false.

The general syntax of the do...while loop is:

```
do
{
    // statements to execute
} while (condition);
```

Key characteristics of the do...while loop:

- • The loop body executes before checking the condition.
- • If the condition is true, the loop body executes again.
- • The process repeats until the condition becomes false.
- • The loop body always executes at least once.

The following table summarizes the differences between while and do...while loops:

Feature	while Loop	do...while Loop
Condition Check	Before execution	After execution
Minimum Executions	Zero (may not execute)	At least one
Syntax	while (condition)	do { } while (condition);
Use Case	When condition is known first	When body must run once

Lab Tasks

Task 1: Numbers 1-10 using while Loop

Write a C++ program that uses a while loop to display numbers from 1 to 10. Display each number on a separate line.

Program Code

```
#include <iostream>
using namespace std;

int main()
{
    int number = 1;

    cout << "Numbers from 1 to 10 using while loop:" << endl;
    cout << "========================================" << endl;

    while (number <= 10)
```

```
    {
        cout << number << endl;
        number++;  // Increment the counter
    }

    return 0;
}
```

Explanation

The program initializes a variable number to 1. The while loop checks if number is less than or equal to 10. If true, it prints the current value and increments it by 1. This process continues until number becomes 11, at which point the condition evaluates to false and the loop terminates.

Task 2: Numbers 1-10 using do...while Loop

Repeat the first task using the do...while loop construct.

Program Code

```
#include <iostream>
using namespace std;

int main()
{
    int number = 1;

    cout << "Numbers from 1 to 10 using do...while loop:" << endl;
    cout << "============================================" << endl;

    do
    {
        cout << number << endl;
        number++;  // Increment the counter
    } while (number <= 10);

    return 0;
```

```
}
```

Explanation

The program uses a do...while loop to achieve the same result as Task 1. The key difference is that the condition is checked after the loop body executes. The loop prints the current number and increments it, then checks if the number is still less than or equal to 10 before deciding whether to continue.

Task 3: Sum using while Loop

Write a C++ program that prompts the user to enter a positive integer. Use a while loop to calculate the sum of all numbers from 1 to the entered integer. Display the calculated sum.

Program Code

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    int sum = 0;
    int counter = 1;

    cout << "Enter a positive integer: ";
    cin >> n;

    // Input validation
    if (n <= 0)
    {
        cout << "Please enter a positive integer." << endl;
        return 1;
    }

    // Calculate sum using while loop
    while (counter <= n)
```

```
    {
        sum = sum + counter;
        counter++;
    }

    cout << "The sum of numbers from 1 to " << n << " is: " << sum << endl;

    return 0;
}
```

Explanation

The program first prompts the user to enter a positive integer and stores it in variable n. It initializes sum to 0 and counter to 1. The while loop iterates as long as counter is less than or equal to n, adding the value of counter to sum in each iteration. After the loop completes, the program displays the calculated sum. The program also includes basic input validation to ensure the user enters a positive number.

Note: The sum of first n natural numbers can also be calculated using the mathematical formula n(n+1)/2, which provides an O(1) solution compared to the O(n) loop approach. However, the loop method is used here to practice while loop implementation.

Outputs and Results

Output for Task 1

```
Numbers from 1 to 10 using while loop:
=========================================
1
2
3
4
5
6
7
```

```
8
9
10
```

Output for Task 2

```
Numbers from 1 to 10 using do...while loop:
============================================
1
2
3
4
5
6
7
8
9
10
```

Output for Task 3

```
Enter a positive integer: 10
The sum of numbers from 1 to 10 is: 55

--- Additional Test Cases ---
Enter a positive integer: 5
The sum of numbers from 1 to 5 is: 15

Enter a positive integer: 100
The sum of numbers from 1 to 100 is: 5050
```

Conclusion

In this laboratory session, we successfully studied and implemented three fundamental loop constructs in C++: the while loop, the nested while loop, and the do...while loop.

The while loop is most suitable when the number of iterations is not known in advance and depends on a condition that is checked before each iteration. The do...while loop is preferred when the loop body must execute at least once, regardless of the initial condition state.

Through the three lab tasks, we demonstrated:

- • Using a while loop to display a sequence of numbers (1 to 10).
- • Using a do...while loop to achieve the same result, understanding the difference in execution flow.
- • Using a while loop with user input to calculate the sum of natural numbers.

Understanding these loop structures is essential for solving problems that require repetitive execution, such as data processing, mathematical calculations, and pattern generation. The concepts learned in this lab form the foundation for more advanced programming techniques involving arrays, file handling, and data structures.

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